
Is There a Universal ‘Sounds Like AI’ Direction? Probing and Steering AI Style in Transformer Residual Streams

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Abstract

1 Detection of AI-generated has become increasingly important as large language
2 models are deployed at scale. Prior work has shown that high-level concepts such as
3 truthfulness, refusal, and sentiment are encoded as linear directions in the residual
4 streams of transformer models. We investigate whether “sounding like AI” — the
5 composite stylistic quality that makes LLM outputs identifiable — is similarly
6 represented, and whether steering along this direction can evade real-world AI
7 detectors. Using contrastive activation addition (CAA) on Qwen2.5-3B, we extract
8 an “AI style direction” from the HC3 dataset. Replicating prior work, we confirm
9 that this direction achieves 97.5% classification accuracy on ChatGPT-sourced
10 text, though 93% of the signal overlaps with text length. After controlling for
11 length, an 85.5% accurate residual style direction remains. Steering along this
12 direction reduces Pangram’s AI probability from 0.79 to 0.02 on ChatGPT-sourced
13 text, demonstrating near-complete evasion of a state-of-the-art detector. We further
14 extend the dataset to include Claude and Gemini responses, finding that these
15 models produce substantially longer outputs (97.5% and 96.1% length overlap
16 respectively), which largely dominates the extracted direction. To address this,
17 we propose a sentence-boundary truncation method that reduces length-direction
18 cosine similarity from 0.944 to 0.002, revealing a genuine cross-source style signal
19 at 95.8% accuracy. However, steering on this length-controlled combined dataset
20 produces inconsistent evasion results, suggesting that a universal model-agnostic
21 AI style direction may not be straightforwardly steerable. Our results highlight
22 both the promise and the limitations of residual stream manipulation as an approach
23 to AI detection evasion.

1 Introduction

24 The proliferation of large language models has intensified an ongoing arms race between AI text
25 generation and AI text detection. Tools such as GPTZero, ZeroGPT, and Pangram attempt to identify
26 machine-generated text by exploiting statistical and stylistic signatures in LLM outputs, while a
27 parallel line of work explores evasion strategies. Existing evasion approaches largely operate at
28 the surface level: recursive paraphrasing Sadasivan et al. [2023], post-hoc humanization, or prompt
29 engineering Mitchell et al. [2023]. These methods treat the model as a black box, modifying outputs
30 after the fact without any understanding of *why* the text sounds AI-generated in the first place. We
31 ask a more fundamental question: is “sounding like AI” encoded as an internal representation within
32 the model itself, and if so, can we manipulate that representation directly to evade detection?

34 A growing body of work in mechanistic interpretability has established that transformer language
35 models linearly encode high-level concepts as directions in their residual streams Marks and Tegmark
36 [2024], Arditì et al. [2024], Konen et al. [2024], and that these directions can be extracted and causally
37 steered via Contrastive Activation Addition Rinsky et al. [2024]. Whether “sounding like AI” — the

38 composite stylistic quality that makes LLM outputs identifiable — constitutes such a direction has
 39 not been directly investigated. Concurrently, Kuznetsov et al. (2025) use Sparse Autoencoders on
 40 residual streams to identify features contributing to AI text detection, finding that style and verbosity
 41 features are the most discriminative. Our work is complementary but distinct: we use the simpler
 42 difference-in-means approach and target *evasion* rather than detection interpretability.

43 We extract an “AI style direction” from Qwen2.5-3B using the HC3 dataset of paired human and
 44 ChatGPT responses, replicating prior work showing 97.5% classification accuracy with 93% overlap
 45 with text length. After controlling for length, a genuine residual style signal persists at 85.5%
 46 accuracy. Critically, steering along this direction reduces Pangram’s AI probability from 0.79 to 0.02,
 47 demonstrating near-complete evasion of a state-of-the-art detector through internal representation
 48 manipulation. We then extend to Claude and Gemini, finding that these models’ substantially greater
 49 verbosity dominates the extracted direction even more severely. To address this, we introduce a
 50 sentence-boundary truncation method that eliminates the length confound by construction, revealing
 51 a genuine cross-source style signal at 95.8% accuracy — but finding that steering on this combined
 52 dataset produces inconsistent evasion results, suggesting that a universal model-agnostic AI style
 53 direction may not be straightforwardly steerable.

54 Our main contributions are:

- 55 • We demonstrate near-complete evasion of Pangram (0.79 $\rightarrow$ 0.02 AI probability) via activa-
 56 tion steering on ChatGPT-sourced text, establishing residual stream manipulation as a viable
 57 attack vector against AI detectors.
- 58 • We extend the analysis to three AI sources (ChatGPT, Claude, Gemini), finding that verbosity
 59 accounts for 93–97.5% of the extracted direction across all sources and is substantially
 60 more severe for Claude and Gemini, suggesting that current AI style directions are largely
 61 verbosity detectors.
- 62 • After controlling for length via sentence-boundary truncation, we find a genuine cross-
 63 source style signal at 95.8% accuracy, but show that this direction does not steer cleanly
 64 across sources — suggesting AI style may be partially model-specific rather than a universal
 65 property of LLM-generated text.

66 2 Background

67 We build directly on a preliminary study Hypogenic-AI [2024] that applied CAA to AI text detection,
 68 which we summarize here as it serves as our baseline.

69 **Dataset.** The study used HC3 Guo et al. [2023], a corpus of paired human and ChatGPT responses to
 70 the same questions across five domains. After filtering to 50–1500 characters, 200 pairs were used
 71 for direction extraction, 50 for validation, and 100 for test. ChatGPT responses are systematically
 72 longer than human responses (mean 914 vs. 446 characters), a confound that substantially affects the
 73 analysis.

74 **Method.** For each layer ℓ of Qwen 2.5-3B, a difference-in-means direction is computed between AI
 75 and human last-token residual stream activations. This direction serves both as a linear classifier and
 76 as a CAA steering vector during generation.

77 **Findings.** The direction at Layer 21 achieves 97.5% test accuracy (AUC = 0.999), but has 0.93 cosine
 78 similarity with a text length direction — it is almost entirely a verbosity detector. After projecting
 79 out length, a residual style direction still achieves 85.5% accuracy at Layer 33. Causal steering
 80 qualitatively shifts output style but with modest effect: an LLM judge found only a ~ 1 point shift on
 81 a 7-point scale, attributed to the base model’s high AI-likeness floor.

82 3 Related Work

83 **Linear representations in residual streams.** A growing body of work in mechanistic interpretability
 84 has shown that transformer language models encode high-level concepts as linear directions in their
 85 residual streams. Marks and Tegmark Marks and Tegmark [2024] provide foundational evidence,
 86 demonstrating that truth and falsehood are linearly represented and extractable via difference-in-
 87 means probing — the core technique we employ — with causal verification via surgical intervention

in the forward pass. Arditi et al. [2024] extend this to behavioral properties, showing that refusal in safety-tuned models is mediated by a single direction whose ablation disables refusal and whose addition induces it on harmless inputs. Rimsky et al. [2024] introduce Contrastive Activation Addition (CAA), the methodology directly underlying our work, which extracts steering vectors by averaging residual stream differences between contrastive pairs and applies them at inference time with a tunable coefficient. Konen et al. [2024] further demonstrate that style properties — sentiment, emotion, writing style — can be extracted and steered via activation engineering without any training. Our work asks whether “sounding like AI” is similarly encoded, extending this line of research to a stylistic property with direct practical implications for detection evasion.

AI text detection and evasion. Mitchell et al. [2023] introduce DetectGPT, a zero-shot detector exploiting the observation that LLM-generated text occupies negative curvature regions of a model’s log probability function. Sadasivan et al. [2023] stress-test the robustness of a broad range of detectors — including watermarking, neural network-based, and zero-shot approaches — showing empirically and theoretically that existing methods are unreliable under paraphrasing attacks. Our approach is fundamentally different from such surface-level evasion: rather than modifying outputs after generation, we intervene directly on the model’s internal style representation during the forward pass.

Residual stream features for AI text detection. Most closely related to our work, Kuznetsov et al. [2025] apply Sparse Autoencoders (SAEs) to Gemma-2-2b’s residual stream to identify features contributing to AI text detection, finding that style and verbosity features are the most discriminative. They also find, consistent with our results, that text length is a major confounding factor. Our work differs in three key ways: we use the simpler difference-in-means approach rather than SAEs; we target evasion rather than detection interpretability; and we evaluate against real-world detectors including Pangram.

4 Methodology

Our methodology extends the preliminary study’s pipeline along three axes: replacing the LLM judge with real AI detectors, testing the instruct variant of the base model, and building a multi-source dataset. We describe each in turn.

4.1 Models

We run all experiments on **Qwen2.5-3B** (base) and **Qwen2.5-3B-Instruct**, loaded in float16 precision. Activations are extracted from the last-token position of the residual stream at all 37 layer outputs (embedding layer + 36 transformer layers), with a maximum input length of 512 tokens.

4.2 Direction Extraction and Classification

We follow the CAA methodology of the preliminary study exactly. For each layer ℓ , the AI direction is computed as the difference-in-means over the training split:

$$\mathbf{d}_\ell = \frac{1}{N} \sum_{i=1}^N \mathbf{a}_\ell^{(\text{AI}, i)} - \frac{1}{N} \sum_{i=1}^N \mathbf{a}_\ell^{(\text{human}, i)}$$

Classification is performed by projecting held-out activations onto the normalized direction and thresholding at the class midpoint. The best layer is selected by validation accuracy. We also compute length-orthogonal directions by projecting out a separately computed length direction (derived from a median split on character count) and sweeping across all layers for the best length-controlled accuracy.

4.3 Multi-Source Dataset

To test whether the AI direction generalizes beyond ChatGPT, we collected responses from **Claude** (claude-sonnet-4-5) and **Gemini** (gemini-2.5-flash) to the same HC3 questions via OpenRouter, yielding approximately 350 pairs per source. We removed the upper character length filter

from the preliminary study, as Claude and Gemini responses are substantially more verbose than ChatGPT (mean lengths of 1,245 and 1,741 characters respectively, vs. 939 for ChatGPT). We run the direction-finding pipeline separately for each source, and also construct a pooled dataset combining all three sources (704 pairs after filtering: 329 ChatGPT, 288 Claude, 87 Gemini) to find a shared AI direction with the human side held constant.

Source	Pairs	Human mean length (chars)	AI mean length (chars)
ChatGPT	350	506	939
Claude	350	506	1,245
Gemini	347	508	1,741
Combined	704	525	1,052
Combined (truncated)	1,031	513	501

Table 1: Dataset statistics per source. Human mean lengths are nearly identical across sources since all draw from the same HC3 questions; the large variation in AI mean length reflects differences in verbosity across models. The combined dataset (704 pairs) applies stricter filtering than the individual pipelines, disproportionately reducing Gemini pairs (329 ChatGPT, 288 Claude, 87 Gemini). The truncated combined dataset retains all 1,031 pairs (345/344/342 per source) by sentence-level truncation of AI responses to match human length, nearly eliminating the length gap.

4.4 AI Detector Evaluation

Rather than using an LLM judge, we evaluate steered outputs against three commercial detectors: **Pangram** (v3 API, `ai_assistance_score`), **GPTZero** (`completely_generated_prob`), and **ZeroGPT** (`fakePercentage/100`). All three return a scalar probability in $[0, 1]$. GPTZero was evaluated manually via web interface due to API cost. We evaluate five steering multipliers (two negative, baseline, two positive) across five fixed prompts, scoring each generated output independently.

4.5 Prompting Baseline

To assess whether instruction prompting achieves comparable evasion to activation steering, we prepend one of three “write like a human” instructions to each prompt and score the outputs with Pangram. We run this on both the base and instruct models: the base model serves as a sanity check (Qwen2.5-3B does not reliably follow instructions), while the instruct model provides a fairer comparison against CAA steering.

4.6 Causal Steering

Steering is implemented via a forward hook registered on the target transformer layer. At each forward pass during generation, the hook adds $\alpha \cdot \hat{\mathbf{d}}_\ell$ to the residual stream at every token position, where α is the steering multiplier and $\hat{\mathbf{d}}_\ell$ is the normalized direction scaled by its raw magnitude. Generation uses temperature 0.7, top-p 0.9, and a maximum of 150 new tokens.

5 Experiments and Results

5.1 Direction Finding: Base Model Replication

We first replicate the preliminary study’s direction-finding pipeline on the HC3 ChatGPT dataset using Qwen2.5-3B. Figure 1 shows classification performance and PCA cluster separation across all 37 layers. The direction-finding results closely match the preliminary study: the best layer by validation accuracy is Layer 21, with test accuracy of 97.5% (AUC = 0.999, 95% CI: [0.950, 0.995]). Cross-layer direction consistency is high (mean adjacent cosine similarity: 0.874), confirming that the direction stabilizes early and remains stable throughout the network. One notable feature is a consistent dip in classification accuracy around layers 8–10 before recovering sharply at layer 11, visible in both the accuracy and silhouette plots; the cause of this is unclear.

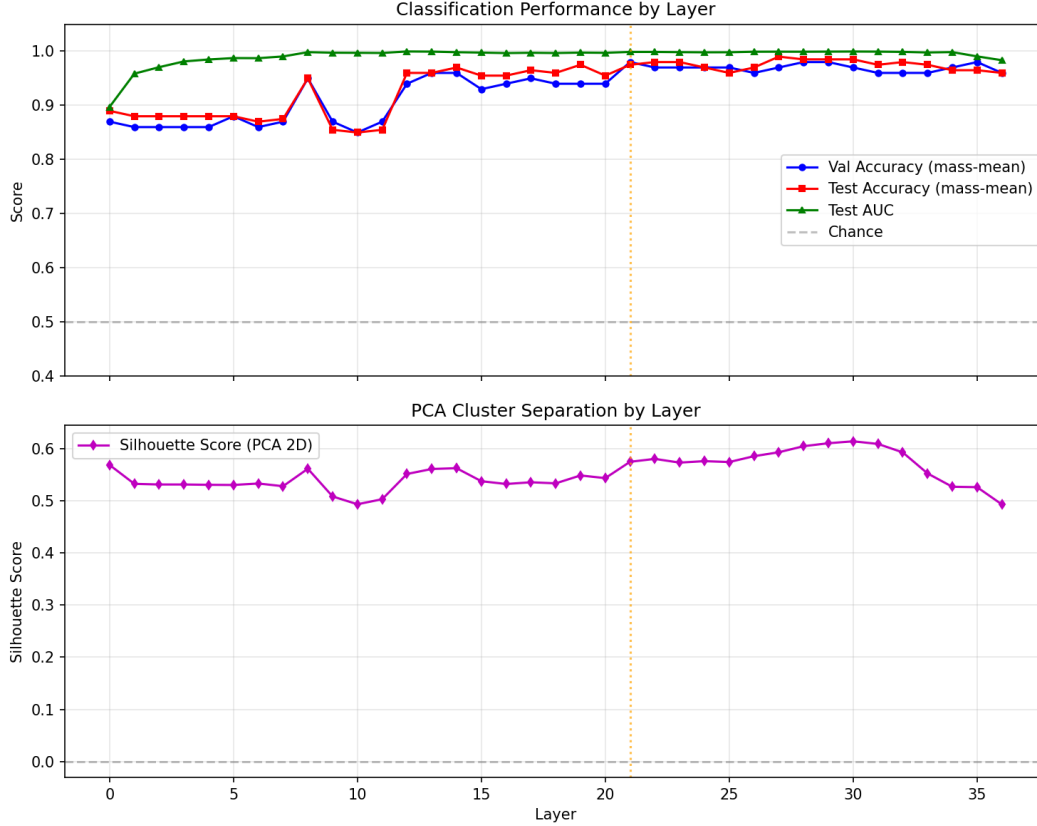


Figure 1: Classification performance (top) and PCA cluster separation (bottom) by layer for Qwen2.5-3B on the HC3 ChatGPT dataset. The orange dotted line marks the best layer (Layer 21). A consistent dip around layers 8–10 is visible in both metrics.

165 The length confound is substantial: cosine similarity between the AI direction and the length direction
 166 is 0.930, and accuracy drops from 97.5% to 82% after projecting out length. The best length-
 167 orthogonal direction is found at Layer 33 with 85.5% accuracy, consistent with the preliminary
 168 study.

169 5.2 Detector Evasion: Base Model

170 Replacing the LLM judge with Pangram yields a substantially stronger evasion result than the
 171 preliminary study suggested. Figure 2 shows Pangram AI probability across five steering multipliers.
 172 At the most human-steered condition (multiplier -33), Pangram assigns only 2% AI probability
 173 — effectively fooled. The baseline (multiplier 0) already scores 0.79, confirming that Qwen2.5-3B
 174 outputs are distinctly AI-sounding even without steering.

175 5.3 Instruct Model

176 Running the same pipeline on Qwen2.5-3B-Instruct yields a different best layer (Layer 24, test
 177 accuracy 97.5%, AUC = 0.998) and a different best length-orthogonal layer (Layer 19, 86% accuracy).
 178 Despite the instruct model having a more pronounced AI style — baseline Pangram score of 0.99 vs.
 179 0.79 for the base model — steering is substantially less effective. The most human-steered condition
 180 only reaches 0.44, compared to 0.02 for the base model. The more entrenched AI style of the instruct
 181 model appears to make it harder to steer away from, rather than providing a larger dynamic range as
 182 initially hypothesized.

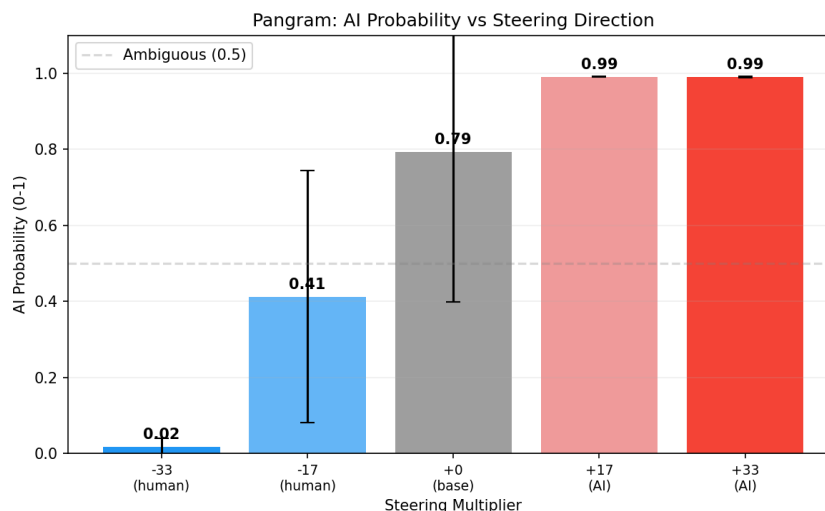


Figure 2: Pangram AI probability by steering multiplier for Qwen2.5-3B steered with the ChatGPT-derived AI direction. Negative multipliers steer toward human-like text; positive multipliers steer toward AI-like text. Error bars show standard deviation across 5 prompts.

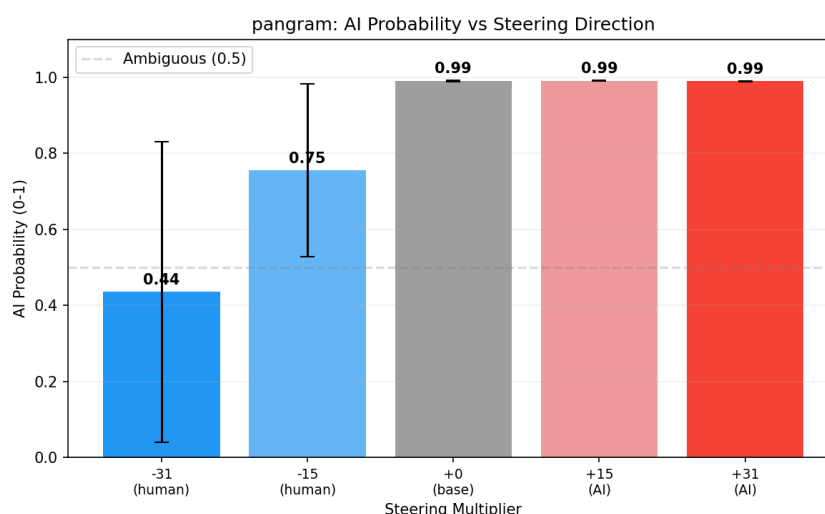


Figure 3: Pangram AI probability by steering multiplier for Qwen2.5-3B-Instruct. Compared to the base model (Figure 2), the baseline is higher (0.99 vs. 0.79) and the most human-steered condition reaches only 0.44 vs. 0.02, suggesting the instruct model’s stronger AI style is harder to override via activation steering.

183 5.4 Does the AI Direction Generalize Across Sources?

184 Table 2 shows direction-finding results across all three sources. All three achieve high classification
185 accuracy, but the length confound is even more severe for Claude and Gemini than for ChatGPT.

186 After projecting out length, Claude and Gemini drop to 62–64% accuracy, barely above chance,
187 compared to ChatGPT’s 80.5%. The within-class correlations tell the same story: Gemini’s correlation
188 between AI text length and projection onto the AI direction is essentially 0 (0.004), meaning Gemini
189 responses are uniformly long regardless of the question. Claude is similar (0.060). This suggests that
190 for Claude and Gemini, the learned direction is almost entirely a verbosity detector with little residual
191 style signal.

Source	Best Layer	Test Acc	Test AUC	Cos sim w/ length	Acc (orig)	Acc (no length)
ChatGPT	22	98.5%	1.000	0.938	98.5%	80.5%
Claude	28	98.0%	0.999	0.975	98.0%	62.0%
Gemini	35	97.4%	0.996	0.961	97.4%	64.4%

Table 2: Direction-finding results per source. The length confound is severe across all sources, but especially for Claude and Gemini, whose length-orthogonal directions barely exceed chance.

Figure 4 shows Pangram evasion results when steering with each source’s own direction. ChatGPT’s direction gives the cleanest evasion (dropping to 0.19 at most human), while Claude and Gemini directions are far less effective (0.75 and 0.85 respectively). The baseline scores for Claude and Gemini are both 0.99, reflecting their high verbosity. One anomaly is that Gemini’s “Most AI” condition actually drops to 0.84, lower than its baseline — likely an artifact of incoherent outputs at high multipliers being misclassified as human by Pangram.

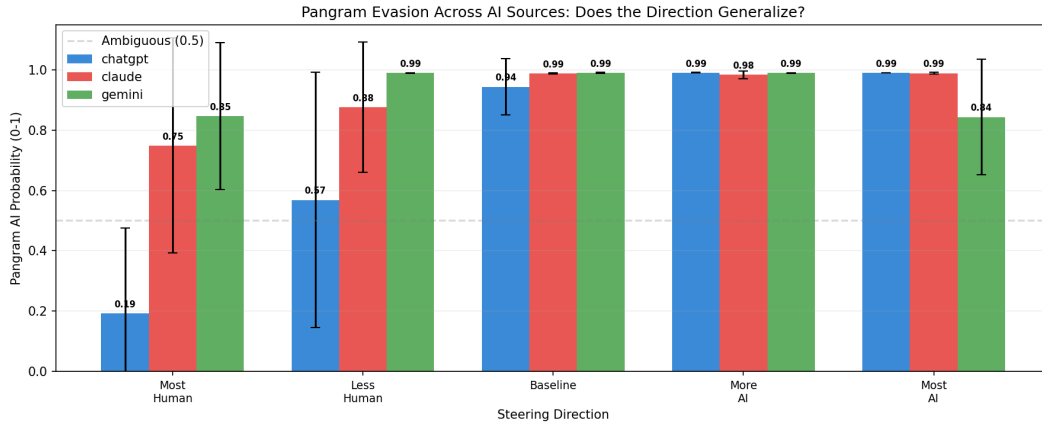


Figure 4: Pangram AI probability by steering direction and source. Each source is steered with its own best-layer direction. ChatGPT’s direction produces the cleanest evasion; Claude and Gemini directions show little effect, consistent with their directions being dominated by length.

5.5 Pooled Dataset

Combining all three sources into a pooled dataset (704 pairs) and finding a single shared direction gives a best layer of 33 with 97.1% test accuracy (AUC = 0.995). The length confound remains strong (cosine similarity 0.944), and accuracy drops from 97.1% to 70.1% after removing length — worse than ChatGPT alone but better than Claude or Gemini individually. Steering on the pooled direction produces a reversed result: negative multipliers (meant to steer human) score *higher* on Pangram than positive multipliers, the opposite of what the ChatGPT-only pipeline found. Inspecting individual outputs reveals that the effect is driven by a few outlier prompts dropping dramatically (e.g. to 0.005–0.067) while most prompts remain near 0.99 regardless of multiplier. We attribute the direction flip to the pooled “AI” centroid being pulled differently when mixing three sources with very different length distributions, potentially causing the mass-mean direction to point in the opposite orientation compared to the ChatGPT-only direction.

5.6 Truncation Experiment

To directly test whether a genuine style signal exists independent of length, we truncate AI responses sentence-by-sentence to match human response lengths (Section 4). This reduces the cosine similarity between the AI direction and the length direction from 0.944 to 0.002, and accuracy before and after removing length is identical (95.8%), confirming that the learned direction is now capturing style rather than verbosity. Direction finding on the truncated dataset gives a best layer of 29 with 95.8% test accuracy (AUC = 0.996, 95% CI: [0.935, 0.977]).

217 However, steering on the truncated dataset remains reversed and noisy (Figure 5), with the same
 218 pattern of a few outlier prompts driving apparent evasion while most prompts show no movement.
 219 This suggests the direction flip is a property of the pooled multi-source direction rather than an artifact
 220 of the length confound.

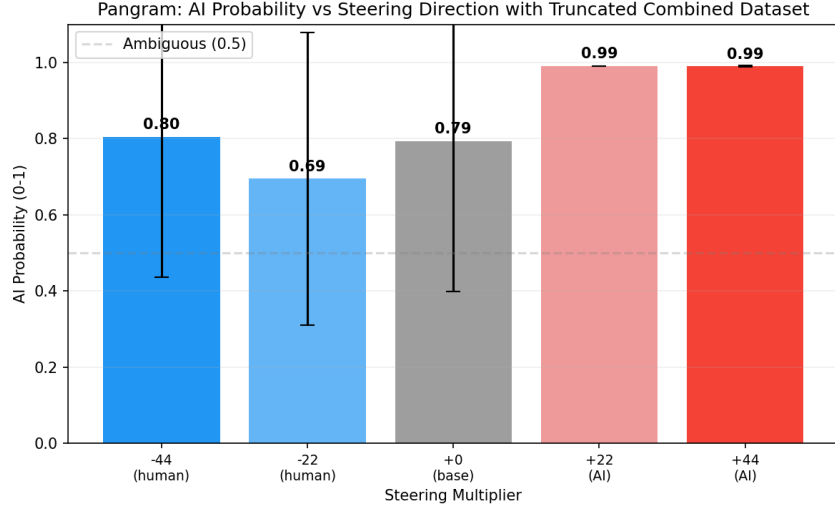


Figure 5: Pangram AI probability by steering multiplier on the truncated pooled dataset. Despite the length confound being eliminated, steering remains reversed and noisy, with high variance across prompts.

221 5.7 Prompting Baseline

222 Figure 6 shows the comparison between CAA steering and prompting baseline on the instruct model.
 223 All three instruction variants flatline at approximately 0.99 Pangram score, identical to the unsteered
 224 baseline. CAA steering reduces the score to 0.44 at the most human-steered condition, substantially
 225 outperforming prompting. On the base model, prompting similarly has no effect, as expected given
 226 that Qwen2.5-3B does not follow instructions reliably.

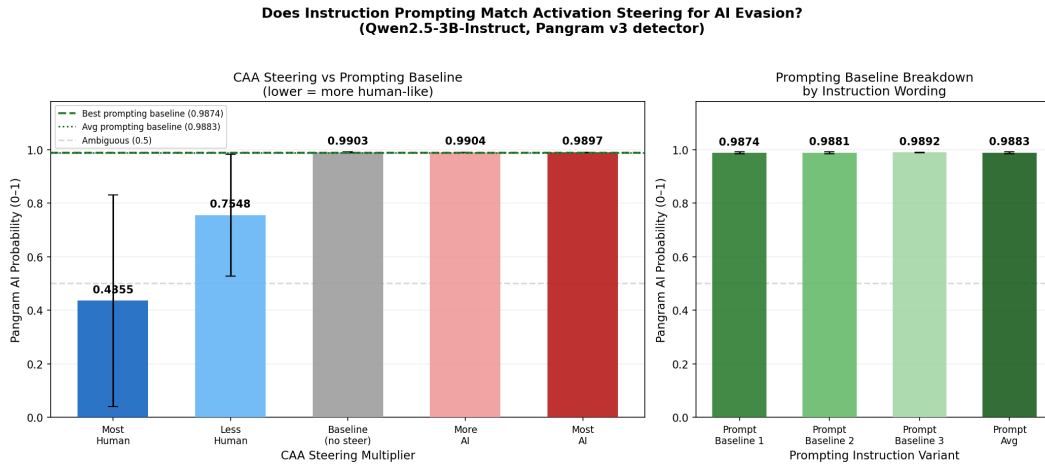


Figure 6: CAA steering vs. prompting baseline on Qwen2.5-3B-Instruct (Pangram v3). Left: steering multipliers with prompting baseline overlaid as dashed lines. Right: breakdown by instruction wording. All prompting variants flatline near 0.99; CAA steering achieves 0.44 at the most human-steered condition.

6 Discussion & Conclusion

The length confound dominates, but a real style signal exists. The most consistent finding across all experiments is that the AI direction is overwhelmingly a verbosity detector. This is most extreme for Claude and Gemini, where over 96% of the learned direction aligns with text length, and length-orthogonal accuracy barely exceeds chance. The truncation experiment provides the clearest evidence that a genuine style signal exists beyond length: after eliminating the length gap, classification accuracy remains at 95.8%, indicating the model encodes real stylistic differences between AI and human text. However, this style signal is harder to exploit for evasion than the length-confounded direction.

A universal AI direction does not straightforwardly exist. The per-source results suggest that the AI direction is source-specific rather than universal. ChatGPT’s direction produces clean, monotonic evasion on Pangram; Claude and Gemini’s directions do not. The pooled direction flips sign relative to the ChatGPT-only direction, and steering with it produces inconsistent results even after controlling for length. This suggests that mixing sources with different verbosity profiles and stylistic quirks disrupts the mass-mean direction in ways that are not easily resolved by simple pooling.

Unsurprisingly, activation steering outperforms prompting The prompting baseline averages around 0.99 on both the base and instruct models with very little variation, while CAA steering achieves 0.02 on the base model and 0.44 on the instruct model. The base model result is somewhat unsurprising since Qwen2.5-3B does not follow instructions reliably, but the instruct model result is more telling: even with a model that can follow instructions, explicit prompting to “write like a human” has no measurable effect on Pangram scores. This suggests that whatever Pangram is detecting is not easily overridden by surface-level stylistic intent, but can be partially overridden by intervening directly on internal representations.

Limitations. Several limitations affect the strength of our conclusions. First, all steering evaluations use only 5 prompts per condition, which is too few to draw robust conclusions — a single incoherent or anomalous output can substantially shift the mean, as seen in the pooled and truncated dataset results. We would have preferred to use more prompts, but scoring each output against AI detectors is expensive: even limiting ourselves primarily to Pangram rather than all three detectors as originally planned, API costs reached approximately \$20 over the course of the project. Second, GPTZero and ZeroGPT results were noisy and non-monotonic, possibly due to the short length of steered outputs (both detectors noted unreliability on inputs less than 250 words); we were unable to draw reliable conclusions from either detector. Finally, all experiments use a single model family (Qwen2.5-3B); generalization to other architectures remains untested.

Conclusion. We extended a preliminary study on mechanistic interpretability of AI-generated text in three directions: multi-source dataset construction, real detector evaluation, and length control. Our results confirm that a linear AI direction exists in Qwen2.5-3B’s residual stream, and that steering with the ChatGPT-derived direction can substantially reduce Pangram detection scores. However, this direction does not generalize cleanly across AI sources, and a pooled multi-source direction fails to produce consistent evasion. The dominant confound throughout is text length, particularly for Claude and Gemini. Taken together, our findings suggest that “sounding like AI” is not a single universal direction in representation space, but a source-specific composite of style and verbosity that varies across model families.

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